Han Peng

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EDUCATION

2014 – **Department of Physics, University of Oxford** DPhil (PhD) Reading in Condensed Matter Physics

2010 - 2014 **University of Science and Technology of China (USTC)** Bachelor of Science in Applied Physics

RESEARCH

2014 – **PhD Project: Study of Electronic Properties of Quantum Materials** **Advisor: Prof Yulin Chen**

Clarendon Laboratory, Department of Physics, University of Oxford

- Using Angular Resolved Photo-Emission Spectroscopy to study electronic properties of quantum materials, including graphene, topological semi-metals and topological insulators
- Developing data analysis system for massive spectrum image data processing. The work includes algorithm optimisation, data visualisation and graphic user interface building

2017 **Summer Research Intern: Neuron Image Registration** **Advisor: Prof Kenneth Harris**

Cortical Processing Laboratory, University College London

- Designed algorithm to register neuronal images with different modality and detection methods
- Developed neuron registration software package for users. Project website: https://git.io/vAjk4

2013 **Research Intern: Study of Jamming Transition in Soft Materials** **Advisor: Prof Ning Xu**

Computational and Theoretical Soft Condensed Matter Physics Laboratory, USTC

- Computationally studied the power law phenomenon in the Jamming transition of a particle system
- Analysed the performance of energy minimization algorithm in different energy potential field

TEACHING

2017 – Tutor in **Mechanics**, Jesus College, University of Oxford

2016 – 2017 Tutor in **Condensed Matter Physics**, Hertford College, University of Oxford

2015 – Demonstrator (Senior Demonstrator since 2017) in **Computational Physics**, University of Oxford

AWARDS AND FUNDING

2014 – Oxford University – China Scholarship Council Scholarship, full scholarship for PhD study

2015 Arthur H. Cooke Memorial Prize awarded by Department of Physics, University of Oxford

2011 National Endeavours Scholarship awarded by Ministry of Education of China

PUBLICATIONS

Total citation: 1272 (Mar 2018). For the updated list, please visit my Google Scholar page: https://goo.gl/vlNvHL

1. **H. Peng**, N. B. M. Schröter, J. B. Yin, H. Wang, T.-F. Chung, H. F. Yang, S. Ekahana, Z. K. Liu, J. Jiang, L. X. Yang, T. Zhang, C. Chen, H. Ni, H. Barinov, Y. P. Chen, Z. F. Liu, H. L. Peng, Y. L. Chen "Substrate Doping Effect and Unusually Large Angle van Hove Singularity Evolution in Twisted Bi- and Multilayer Graphene" ***Advanced Materials (2017)***
2. J. Jiang, Z. K. Liu, Y. Sun, H. F. Yang, R. Rajamathi, Y. P. Qi, L. X. Yang, C. Chen, **H. Peng**, C.-C. Hwang, S. Z. Sun, S.-K. Mo, I. Vobornik, J. Fujii, S. S. P. Parkin, C. Felser, B. H. Yan, Y. L. Chen "Signature of type-II Weyl semimetal phase in MoTe2" ***Nature Communications, 8, 13973 (2017)***
3. Z. K. Liu, L. X. Yang, Y. Sun, T. Zhang, **H. Peng**, H. F. Yang, C. Chen, Y. Zhang, Y. F. Guo, D. Prabhakaran, M. Schmidt, Z. Hussain, S.-K. Mo, C. Felser, B. Yan and Y. L. Chen "Evolution of the Fermi surface of Weyl semimetals in the transition metal pnictide family" ***Nature Materials, 15, 27 (2016)***
4. J. B. Yin†, H. Wang†, **H. Peng† (equal contribution)**, Z. J. Tan, L. Liao, L. Lin, X. Sun, A. L. Koh, Y. L. Chen, H. L. Peng, and Z. F. Liu "Selectively enhanced photocurrent generation in twisted bilayer graphene with van Hove singularity" ***Nature Communications, 7, 10699 (2016)***
5. L. Liao, H. Wang, **H. Peng**, J. B. Yin, A. L. Koh, Y. L. Chen, Q. Xie, H. L. Peng, and Z. F. Liu "van Hove Singularity Enhanced Photochemical Reactivity of Twisted Bilayer Graphene" ***Nano Letters, 15, 5585 (2015)***
6. L. X. Yang, Z. K. Liu, Y. Sun, **H. Peng**, H. F. Yang, T. Zhang, B. Zhou, Y. Zhang, Y. F. Guo, M. Rahn, D Prabhakaran, Z. Hussain, S.–K. Mo, C. Felser, B. Yan and Y. L. Chen "Weyl Semimetal Phase in non-Centrosymmetric Compound TaAs" ***Nature Physics, 11, 728 (2015)***
7. Z. K. Liu, J. Jiang, B. Zhou, Z. J. Wang, Y. Zhang, H. M. Weng, D. Prabhakaran, S. -K. Mo, **H. Peng**, P. Dudin, T. Kim, M. Hoesch, Z. Fang, X. Dai, Z. X. Shen, D. L. Feng, Z. Hussain, Y. L. Chen. "A Stable Three-dimensional Topological Dirac Semimetal Cd3As2" ***Nature Materials, 13, 677 (2014)***